

# DUBAI LAND DEPARTMENT

## REQUEST FOR PROPOSAL

RFP No.: DLD-IRETP-2026-001

### Integrated Real Estate Transparency Platform (IRETP)

|                     |                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Issuing Authority:  | Dubai Land Department (DLD)                                                                                                          |
| RFP Version:        | 1.0                                                                                                                                  |
| Issue Date:         | April 2026                                                                                                                           |
| Classification:     | CONFIDENTIAL                                                                                                                         |
| Project Duration:   | 4 calendar months from contract signing (four phased deliveries). Go-live only after successful completion of all mandatory testing. |
| Post-Go-Live:       | 12-month warranty & technical support following final go-live sign-off                                                               |
| Evaluation Weights: | Technical Requirements: 70%   Budget & Cost: 30%                                                                                     |

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# 1. Executive Summary

The Dubai Land Department (DLD) hereby issues this Request for Proposal (RFP) for the design, development, deployment, maintenance, and post-go-live support of the Integrated Real Estate Transparency Platform (IRETP). The primary strategic objective is to advance Dubai's standing in the JLL Global Real Estate Transparency Index (GRETI) and to reinforce investor confidence among international financial institutions, sovereign wealth funds, and global investors.

The platform directly supports the Dubai Economic Agenda D33 and the Smart Dubai Strategy by providing open, accurate, real-time, and interactive real estate data to the public while equipping DLD leadership with powerful internal risk management and developer oversight tools.

**The IRETP comprises two integrated pillars:**

- External Public Portal — A world-class, multilingual data portal for international investors, financial institutions, property developers, and the general public, offering interactive dashboards, a GIS map, an AI-powered data agent, self-service analytics, and open-data export capabilities.
- Internal Management Platform — A secure, role-based platform for DLD staff and leadership, encompassing a developer performance rating engine, Escrow account monitoring, an Early Warning Risk System (EWRS), and multi-level leadership alert workflows.

This RFP does not prescribe specific technology brands, frameworks, programming languages, or hosting providers. Vendors are expected to propose the solution architecture, technology stack, and hosting model that best meets the requirements described herein. Proposals are evaluated on 70% technical merit and 30% budget and total cost of ownership.

Delivery is structured across four phases, all of which must be completed within 3 calendar months from the date of contract signing. Go-live of any phase is conditional on the successful completion of all mandatory testing and acceptance criteria for that phase — no phase may go live before its tests are fully passed. Following the go-live of the final phase, the vendor must provide a 12-month warranty and technical support period as part of this contract.

## 1.1 Platform Objectives

| Objective               | Description                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| GRETI Score Improvement | Directly address the six JLL GRETI sub-indices through open data, AI integration, and market transparency measures.    |
| Investor Confidence     | Provide institutionally credible, machine-readable, and auditable real estate data for international markets.          |
| Data Accuracy           | Ensure all published data undergoes rigorous analytics assessment against DLD source systems before and after go-live. |

| Objective              | Description                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developer Governance   | Establish a transparent, data-driven developer accountability framework accessible to DLD leadership.                                                   |
| Market Risk Management | Equip DLD with proactive risk detection tools identifying project and market stress indicators before escalation.                                       |
| Sustained Operations   | Ensure platform stability, security, and timely defect resolution through a mandatory 12-month warranty and technical support period following go-live. |

## 2. Project Overview & Strategic Context

### 2.1 Background

The JLL Global Real Estate Transparency Index (GRET<sup>®</sup>I), published biennially, is the primary reference used by institutional investors, lenders, and multinational corporations to evaluate real estate market quality globally. The 2024 edition assessed 256 individual indicators across 89 countries and 151 cities. Markets in the 'Highly Transparent' tier attract over 80% of global direct commercial real estate investment. Dubai has demonstrated consistent improvement, and this platform is designed to accelerate that progress.

### 2.2 GRET<sup>®</sup>I Sub-Index Gap Analysis

| GRET <sup>®</sup> I Sub-Index           | Identified Gap                                            | How IRETP Addresses It                                                                            |
|-----------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Investment Performance Measurement      | Limited public price indices and rental yield data        | Real-time price-per-sqft index, rental yield heatmaps, and historical transaction charts by zone  |
| Market Fundamentals & Data Availability | Fragmented, non-machine-readable data                     | Open Data API, Excel/CSV/PDF exports, structured data by property type, zone, and period          |
| Transaction Process Transparency        | Partial visibility into off-plan and gifting transactions | Full transaction registry: sales, gifts, mortgages, auctions — publicly searchable and exportable |
| Technology & AI Integration             | No conversational AI or self-service analytics            | AI Agent with RAG architecture, Slice & Dice analytics engine, and in-chat dashboard generation   |
| Sustainability / ESG                    | No publicly accessible green-building data layer          | ESG module displaying LEED/Estidama certification data (Phase 4)                                  |
| Governance of Listed Vehicles           | Developer risk data not publicly accessible               | Public Developer Scorecard and internal Developer Rating Platform (Phase 3)                       |

### 2.3 Project Scope Summary

1. External Public Real Estate Data Portal — investor-facing, bilingual (Phase 1).
2. Real Estate AI Agent — deployed on both External and Internal platforms (Phase 1 onwards).
3. Interactive Analytics & Slice-and-Dice Engine (Phase 1).
4. Internal Platform — Early Warning Risk System (EWRS) with multi-level leadership alerts (Phase 2).
5. Internal Platform — Developer Performance & Rating with Escrow Monitoring (Phase 3).
6. External Portal — Extended Languages (6 additional languages, Phase 4).
7. Headless Content Management System (CMS) for non-technical content updates (Phase 1).
8. Open Data API & Developer Portal (Phase 1).

9. DLD Data Understanding, Integration, and Analytics Assessment — mandatory pre-publication data validation (Phase 1, ongoing).
10. Infrastructure, Security Architecture, and DESC Compliance Framework (all phases).
11. 12-Month Warranty & Technical Support following final phase go-live.

### 3. Functional Requirements — External Public Portal

#### 3.1 Content Management System (CMS)

The platform must allow DLD non-technical staff to update, add, or remove page content — text, images, data annotations, banners, and calls-to-action — through a visual administration interface, without any developer involvement or re-deployment.

| Req. ID | Requirement                          | Specification                                                                                                                                                                                                                                                         | Acceptance Criterion                                                                                                                           |
|---------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-001  | Headless CMS — Non-Technical Editing | API-first headless CMS proposed by the vendor. Full RBAC within CMS. Supports content types: rich text, images, structured data tables, embedded charts, map annotations, hyperlinks, and banners. Must support Arabic (RTL) and English from a single content model. | Any main portal page fully updated by a non-developer DLD staff member in under 10 minutes. Content changes live within 30 seconds of publish. |
| FR-002  | Staging & Publication Workflow       | Must support staging → preview → production workflow. Preview link shareable with DLD senior management for approval before publishing. Version history retained for at least 12 months with rollback capability.                                                     | Rollback to any prior version within 2 clicks. Management can preview changes on staging URL before approval.                                  |

#### 3.2 Homepage — Market Overview Dashboard

| Req. ID | Feature                   | Specification                                                                                                                                                                                                                                                                                                                           | Acceptance Criterion                                                                         |
|---------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| FR-003  | Real-Time KPI Cards       | Display live KPIs: (a) Total transactions this month (count and AED value), (b) Average price per sqft — Dubai-wide, (c) Total completed units YTD, (d) Active off-plan projects count, (e) Average residential rental yield %. Each card shows a trend arrow vs. prior period (MoM and YoY). Cards must be CMS-configurable.           | KPIs refresh every 15 minutes. Data lag from DLD source systems must not exceed 15 minutes.  |
| FR-004  | Interactive Market Charts | Minimum four interactive charts: (1) Monthly transaction volume — line chart, 24-month history. (2) Transaction type breakdown — donut chart. (3) Top 5 most-active zones — bar chart. (4) Price-per-sqft trend — dual-axis line chart. All charts support hover tooltips, zoom, and pan. Period filters: Last 6M / 12M / 36M / Custom. | Charts render within 2 seconds after filter change. All chart data traceable to DLD records. |

| Req. ID | Feature                      | Specification                                                                                                                                                                                                                   | Acceptance Criterion                                              |
|---------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| FR-005  | Language & Currency Switcher | Language switcher in global header on all pages. Currency switcher (AED / USD / EUR / GBP / CNY / RUB) applying daily exchange rates from UAE Central Bank API or equivalent. User preference stored in browser and in account. | Language switch takes effect within 1 second with no page reload. |

### 3.3 Transactions Page

| Req. ID | Feature                   | Specification                                                                                                                                                                                                                                                                                                                                                                                             | Acceptance Criterion                                                                                               |
|---------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| FR-006  | Full Transaction Registry | Searchable, filterable table of all recorded transactions. Each row: Transaction Date, Zone/Community, Project Name, Property Type (Apartment / Villa / Townhouse / Plot / Commercial), Transaction Type (Sale / Gift / Mortgage / Auction / Inheritance), Total Area (sqft and sqm), Transaction Value (AED), Price Per Sqft, Financing Method. <b>Minimum 5 full calendar years of historical data.</b> | 5-year data accessible. Table loads within 3 seconds for up to 10,000 rows.                                        |
| FR-007  | Multi-Dimensional Filters | Filters operating simultaneously with real-time update: Date Range, Zone/Community (multi-select), Property Type (multi-select), Transaction Type, Price Range (dual-handle slider), Area Range (sqft), Financing Method. Active filter summary bar shown at all times.                                                                                                                                   | Any filter combination updates results within 2 seconds. Filter state persists on page refresh via URL parameters. |
| FR-008  | Transaction Data Export   | Export full filtered dataset in: (a) Excel (.xlsx) with DLD header, timestamp, and applied filters in a metadata sheet, (b) CSV — raw data, (c) PDF — formatted report with DLD branding and summary statistics. Maximum 50,000 rows per export.                                                                                                                                                          | Export and download begins within 10 seconds for datasets up to 50,000 rows.                                       |

### 3.4 Interactive GIS Map — Zones & Projects

| Req. ID | Feature          | Specification                                                                                                                                                  | Acceptance Criterion                                                |
|---------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| FR-009  | GIS Zone Heatmap | Full-screen interactive GIS map of Dubai with zone-level data overlays. Three switchable heatmap layers: (1) Transaction Activity — intensity reflects monthly | Map initialises within 5 seconds on a 10 Mbps connection. All three |



| Req. ID | Feature                   | Specification                                                                                                                                                                                                                                                                                                                                                                                                        | Acceptance Criterion                                                                          |
|---------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|         |                           | volume, (2) Average Price Per Sqft — colour ramp from low to high, (3) Rental Yield — gross yield % per zone. Zone boundaries from official Dubai Municipality GIS data.                                                                                                                                                                                                                                             | layers switch without page reload.                                                            |
| FR-010  | Zone Detail Panel         | Clicking any zone opens a panel displaying: Zone Name, Total Transactions (last 12 months), Average Sale Price Per Sqft, Average Annual Rental by unit type, Average Gross Rental Yield (%), Price Trend Indicator (up/stable/down with % change), Top 3 active developers, Active off-plan projects count, Completed projects count. A 'View Projects in This Zone' button navigates to the filtered Projects page. | Panel opens within 1 second of zone click. All fields populated from live DLD data.           |
| FR-011  | Project Pins & Clustering | All active projects as interactive map pins, colour-coded by status: Completed, Under Construction, Future/Announced, Stalled. Pins cluster at low zoom levels. Clicking a cluster zooms to expand. Map filter controls update pin visibility in real time.                                                                                                                                                          | All project pins render within 3 seconds. Clustering correct at all zoom levels.              |
| FR-012  | Project Detail Panel      | Clicking a project pin opens a panel with: Project Name, Developer Name and Licence Number (linked to developer profile), Status badge, Completion Percentage (progress bar), Total/Sold/Available Units, Unit Types with average prices, Expected Delivery Date, Location mini-map, Link to official DLD Project Registration Record.                                                                               | All data populated from DLD official records. Developer cannot self-edit any displayed field. |

### 3.5 Price Index & Rental Index Pages

| Req. ID | Feature                         | Specification                                                                                                                                                                                                                                                                                 | Acceptance Criterion                                                         |
|---------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| FR-013  | Price Per Sqft Index            | Interactive index showing average transaction price per sqft across Dubai. Dimensions: property type, transaction status (Off-Plan / Ready), time range (zoomable 10-year trend). Zone comparison — user selects up to 5 zones for an overlaid chart. Quarterly and annual change indicators. | Index data derived exclusively from actual DLD transactions. Updated weekly. |
| FR-014  | Rental Index & Yield Calculator | Residential and commercial rental index per zone: average annual rent by unit type,                                                                                                                                                                                                           | Rental data sourced from registered Ejari records                            |

| Req. ID | Feature | Specification                                                                                                                                                                                                         | Acceptance Criterion                                |
|---------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|         |         | 5-year historical trend, Gross Rental Yield = (Annual Rent / Transaction Price) × 100 per zone and unit type, yield heatmap linked to the GIS map. Segment filters: Residential / Commercial, Short-Term / Long-Term. | or DLD-endorsed data. Updated quarterly at minimum. |

## 4. Interactive Analytics Engine — Slice & Dice

The platform must provide a self-service analytics engine enabling any user — without technical expertise — to construct, explore, and export custom data analyses through a visual guided interface.

| Req. ID | Capability                        | Specification                                                                                                                                                                                                                                                                                     | Acceptance Criterion                                                          |
|---------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| AN-001  | Dimension & Metric Selection      | Users select dimensions (Zone, Developer, Property Type, Transaction Type, Time Period, Construction Status) and metrics (Transaction Count, Total Value AED, Average Price/Sqft, Rental Yield %, Units Count, Completion %). Any combination of up to 4 dimensions and 3 metrics simultaneously. | Any valid combination renders a correct chart within 3 seconds.               |
| AN-002  | Visualization Types               | Results in user-selectable types: Bar, Stacked Bar, Line, Area, Scatter, Donut, Treemap, Data Table, KPI Summary Cards. System recommends the most appropriate type.                                                                                                                              | All 9 chart types render correctly for all valid data combinations.           |
| AN-003  | Saved Views & Personal Dashboards | Users can save any configured analysis view. Registered users can build a personal dashboard from up to 12 saved views in a drag-and-drop grid. Dashboards can be shared via a secure, read-only link.                                                                                            | Saved view restores all filter states, chart type, and dimensions accurately. |
| AN-004  | Full-Dataset Export               | Export the complete filtered dataset in: Excel (.xlsx) with summary statistics sheet, CSV (raw data), PDF (formatted report with DLD letterhead and embedded chart), JSON (for API users). Exports include: date generated, filters applied, data source reference, and disclaimer note.          | Export of 50,000-row dataset completes within 15 seconds.                     |
| AN-005  | Zone Comparison Module            | Dedicated comparison view: up to 5 zones compared across any metrics over a selected time period, displayed as side-by-side bar charts, overlaid line charts, and a sortable comparison table.                                                                                                    | Comparison of 5 zones with 3 metrics renders within 4 seconds.                |
| AN-006  | Shareable Analysis Links          | Every analysis configuration generates a unique shareable URL encoding all selected dimensions, metrics, filters, chart type, and time period. Links valid for a minimum of 12 months.                                                                                                            | Shared link restores analysis state with 100% fidelity.                       |

## 5. Real Estate AI Agent Requirements

### 5.1 Scope & Critical Constraint

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRITICAL CONSTRAINT — NO INVESTMENT ADVISORY The AI Agent is strictly prohibited from providing personalised investment advice, recommending specific properties or developers for purchase, making price forecasts presented as facts, or any other form of financial or legal advisory. The Agent's role is exclusively to retrieve, present, and analyse data from the DLD database and to guide users to official DLD services. This constraint must be enforced at the system-prompt level and validated during UAT. |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.2 Core Agent Capabilities

| Req. ID | Capability                    | Specification                                                                                                                                                                                                                                                                       | Constraints & Standards                                                                                                                                |
|---------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI-001  | Natural Language Data Queries | The agent must accurately respond to any natural-language question about DLD database data in both English and Arabic. All answers must cite the source data and time period. The agent must acknowledge when data is unavailable rather than fabricating an answer.                | Accuracy benchmark: greater than 90% correct answers on a standardised 100-question DLD data test set. Source citation mandatory for every data point. |
| AI-002  | In-Chat Dashboard Generation  | Upon user request, the agent must generate fully interactive charts and dashboard widgets directly within the chat window. Generated charts must support hover tooltips, zoom, and axis labels, accompanied by a brief textual data summary.                                        | Charts generated from actual DLD data only. Maximum 30-second generation time.                                                                         |
| AI-003  | In-Chat Download              | After generating any chart, table, or analysis, the agent must display a download button offering: PDF (chart + data table + DLD branding), Excel (.xlsx), PNG (chart image only), CSV (raw data). Files include: DLD logo, timestamp, query parameters, and data source statement. | Download button appears within 2 seconds of chart generation. All four formats function correctly.                                                     |
| AI-004  | Deep Data Analysis            | Support complex analytical queries: correlation analysis, anomaly detection in transaction volumes, multi-year trend analysis, developer performance comparison. The agent must explain its methodology in plain language.                                                          | Complex queries answered within 15 seconds. Trend analysis clearly labelled as historical, not predictive.                                             |
| AI-005  | DLD Service Navigation        | Guide users to official DLD services with step-by-step instructions: Property Registration, RERA NOC, Developer licence verification, Ejari registration,                                                                                                                           | Service information reviewed and approved by DLD operations team before go-live.                                                                       |

| Req. ID | Capability             | Specification                                                                                                                                                                                                         | Constraints & Standards                                                                            |
|---------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|         |                        | Valuation services — with links to official DLD service pages.                                                                                                                                                        |                                                                                                    |
| AI-006  | Session Context Memory | Agent maintains complete context within a session. For authenticated users: optional cross-session memory of zone preferences and frequent query topics. Memory opt-in/opt-out user-controlled from account settings. | Agent resolves context correctly in more than 95% of multi-turn conversations in the UAT test set. |
| AI-007  | Multilingual Support   | Phase 1: Full conversational capability in English and Arabic. Phase 4: Basic query handling in Chinese (Simplified), Russian, Urdu, French, Hindi, and German with graceful English fallback.                        | Arabic NLP accuracy within 5% of English accuracy on the same standardised test set.               |

### 5.3 AI Model Architecture — Multi-Model Approach

DLD does not prescribe a specific AI model, vendor, or provider. The platform must support a multi-model AI architecture that gives DLD the flexibility to use more than one AI model — either simultaneously for different tasks, or switchable by the platform administrator without requiring redevelopment. The vendor must propose and justify their multi-model architecture in the proposal.

The following requirements apply to the AI architecture:

| AI Architecture Requirement | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Model Support         | The platform must support the use of more than one AI language model. The vendor must design an orchestration layer (e.g., an AI routing or gateway component) that sits between the AI Agent and the underlying models, enabling DLD to: (a) use different models for different task types (e.g., one model for data queries, another for service navigation), (b) switch the active model from the administration interface without code changes or redeployment, and (c) add a new AI model provider in the future by configuration, not by rebuilding the AI layer. |
| Acceptable Model Options    | The AI orchestration layer must be compatible with, and the proposal must demonstrate how it will support, a range of model types including but not limited to: managed AI API services (such as OpenAI API, Anthropic API, Google Gemini API, or equivalent), self-hosted open-source LLMs (such as Llama, Mistral, or equivalent models hosted within the UAE), and custom fine-tuned models trained on DLD domain data. The vendor may propose any combination of these. DLD does not mandate a specific model or provider.                                          |
| Mandatory RAG Layer         | A Retrieval-Augmented Generation (RAG) layer is mandatory regardless of which AI model is used. The RAG layer must index the DLD transaction database, project registry, developer profiles, RERA regulations, and DLD service procedures, and must ground every AI Agent response in retrieved data                                                                                                                                                                                                                                                                    |

| AI Architecture Requirement    | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | before generation. The RAG layer must be model-agnostic — it must function correctly regardless of which underlying LLM is active.                                                                                                                                                                                                                                                                                                                 |
| Model Fallback & Redundancy    | If the primary AI model is unavailable (API outage, service degradation), the system must automatically fall back to a secondary model without user-visible disruption. The vendor must describe the fallback model and the expected impact on response quality during fallback. The AI Agent must never display a raw API error to the end user.                                                                                                  |
| UAE Data Residency             | All AI inference and data processing must occur within the United Arab Emirates, regardless of which AI model is active. If using a managed AI API service, the vendor must confirm that the provider offers a UAE or Gulf-region inference endpoint and that data does not transit outside the UAE. If no UAE endpoint is available for a proposed model, the vendor must propose a compliant alternative.                                        |
| Custom Fine-Tuning Option      | The vendor must indicate whether they propose custom fine-tuning of one or more models on DLD domain data (real estate terminology, RERA procedures, zone nomenclature, transaction classification). If fine-tuning is proposed, the vendor must describe the training data approach, the fine-tuning schedule, and how model accuracy will be validated post-fine-tuning. Fine-tuning is not mandatory but will be evaluated as a differentiator. |
| Model Performance Transparency | The administration interface must display, for each configured model: the model name and version, the current active/inactive status, the average response latency over the last 7 days, and any active fallback conditions. DLD administrators must be able to view this information without vendor assistance.                                                                                                                                   |

## 6. Investor Notification & Alert System

### 6.1 External Investor Alert Types

| Alert Type                      | Trigger Condition (User-Configurable)                                                                                    | Delivery Channels                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Price Movement Alert            | Average price per sqft in a selected zone changes by more than X% (user sets threshold 1%–25%) within a selected period. | Email (HTML) + In-Platform Notification    |
| New Project Launch              | A new project is registered in DLD in a zone or by a developer selected as a preference.                                 | Email + SMS (optional) + Push notification |
| Watchlist Project Status Change | Any project in the investor's Watchlist changes construction status, completion percentage, or delivery date.            | Email + In-Platform + SMS (optional)       |
| Rental Yield Threshold          | Average gross rental yield in a selected zone exceeds or falls below a user-defined percentage.                          | Email + In-Platform Notification           |
| Periodic Market Digest          | Weekly or monthly summary of market movements in the investor's zones of interest.                                       | Email (HTML with embedded charts)          |
| Regulation / Policy Update      | A DLD regulation change or RERA directive relating to selected property types or zones.                                  | Email + In-Platform Notification           |

### 6.2 Notification Delivery Standards

- Email: professional HTML template with DLD branding, embedded chart for price/yield alerts, clear call-to-action link, and unsubscribe link.
- SMS: plain-text under 160 characters, including the critical metric value and a shortened link.
- In-Platform Notification Centre: persistent bell icon in header, unread count badge, notification list with read/unread states, mark-all-as-read function.
- Delivery SLA: email within 5 minutes of trigger; SMS within 3 minutes; in-platform notification instantaneous.

## 7. Multilingual Requirements

| Platform                   | Languages Required                                       | Coverage Scope                                                       | Delivery Phase |
|----------------------------|----------------------------------------------------------|----------------------------------------------------------------------|----------------|
| Internal Platform          | English, Arabic                                          | Full UI, all content, AI Agent, alert messages                       | Phase 1        |
| External Portal — Core     | English, Arabic                                          | Full UI, all content, AI Agent (full NLP), export labels             | Phase 1        |
| External Portal — Extended | Simplified Chinese, Russian, Urdu, French, Hindi, German | Full UI translation + static content + AI Agent basic query handling | Phase 4        |

### 7.1 Technical Localisation Standards

- RTL (Right-to-Left) layout must be fully applied for Arabic and Urdu — including layout mirroring of navigation, icon placement, chart axis alignment, and table column order.
- All text stored in Unicode/UTF-8. No hardcoded strings in the codebase — all translatable strings managed through i18n resource files or CMS locale fields.
- Chart labels, axis titles, data tooltips, export headers, and PDF reports must all respect the active language.
- Arabic translation of official zone names, project names, and developer names must be validated by DLD's official records before go-live.
- Language preference stored per user account and in browser storage. Language can be changed without losing current page state.



## 8. Internal Platform — Early Warning Risk System (EWRs)

### 8.1 Risk Indicators & Thresholds

| Risk Indicator                    | Default Trigger Threshold                                                              | Risk Level                                  | Escalation Path                        |
|-----------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------|
| Project Delivery Delay — Warning  | Declared delivery date exceeded by more than 6 months with less than 80% completion    | Warning                                     | Level 1 to Level 2                     |
| Project Delivery Delay — Critical | Declared delivery date exceeded by more than 12 months with less than 90% completion   | High Risk                                   | Levels 1, 2, and 3                     |
| Escrow Shortfall — Warning        | Escrow balance below 80% of required minimum                                           | Warning                                     | Level 1 to Level 2                     |
| Escrow Shortfall — Critical       | Escrow balance below 60% of required minimum                                           | High Risk                                   | Levels 1, 2, 3, and 4                  |
| Construction Activity Suspension  | No construction progress update registered for more than 30 consecutive days           | Warning (30+ days) / High Risk (60+ days)   | Level 2 at 30 days; Level 3 at 60 days |
| Sharp Transaction Volume Decline  | Monthly transactions in a zone drop more than 40% vs. the 12-month rolling average     | Medium                                      | Level 2 (Zone Risk Report)             |
| Developer Score Deterioration     | Developer composite score drops more than 15 points in a single quarter                | Medium to High Risk if score falls below 40 | Level 2 + Developer Review Flag        |
| High-Risk Project Concentration   | Portfolio share of High Risk projects exceeds 30% of all active projects               | High Risk                                   | Level 3 + Executive Summary Report     |
| Price Decline — Zone Level        | Average sale price per sqft in any zone declines more than 15% within a single quarter | Medium                                      | Level 2 (Market Risk Report)           |
| Severe Regulatory Violation       | Developer receives a Critical-category RERA violation                                  | High Risk                                   | Level 3 + Automatic Audit Flag         |

Note: All threshold values are defaults. Authorised DLD administrators must be able to adjust every threshold from the EWRs configuration panel. All changes must be logged with timestamp and approver identity.

### 8.2 Multi-Level Alert Escalation Framework

| Alert Level                 | Recipients                                       | Delivery Channels                                                                      | Response SLA                                                                   |
|-----------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Level 1 — Operational       | Assigned DLD project officer                     | Internal platform notification + email                                                 | Acknowledge within 4 business hours; action plan within 2 business days        |
| Level 2 — Managerial        | Section Manager + Division Director              | Email + SMS + platform notification with priority badge                                | Acknowledge within 2 business hours; escalation decision within 1 business day |
| Level 3 — Senior Leadership | Director General + Deputy Directors General      | Encrypted email + SMS + auto-generated Executive Summary PDF                           | Acknowledge within 1 hour (business hours); within 4 hours (after hours)       |
| Level 4 — Strategic         | Director General + Relevant Regulatory Authority | Formal encrypted email + SMS + automated briefing document + optional robocall trigger | Immediate — no standard SLA: notification is regulatory or crisis level        |

### 8.3 EWRS Dashboard Interface Requirements

- Main Risk Overview Dashboard: Dubai-wide risk heatmap on the GIS map (project pins colour-coded by risk level). Summary KPI bar: total High Risk projects, total Warning projects, projects with Escrow shortfall, projects with construction halt.
- Alert Inbox: chronological log of all generated alerts with status (New / Acknowledged / Resolved / Escalated), assigned owner, timestamps, and action notes.
- Playbook Integration: each alert type links to a pre-defined response SOP with a checklist DLD staff can mark off, creating an audit trail.
- Threshold Configuration Panel: authorised administrators can adjust any threshold value. Changes logged with timestamp and approver identity.

### 8.4 Escrow Account Monitoring

| Req. ID | Feature                    | Specification                                                                                                                                                                                                                                                                                                                                            | Alert Behaviour                                                                                                                  |
|---------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| ESC-001 | Real-Time Escrow Dashboard | Per-project display of: Current Escrow balance (AED), Required minimum balance (calculated from construction completion % × total project cost), Balance adequacy ratio, Total funds received from buyers, Total authorised withdrawals, Remaining construction cost estimate. Status badges: Adequate (≥ 100%), Warning (80–99%), Critical (below 80%). | Automatic alert generated when any project drops to Warning. Immediate escalation to Level 2 when any project drops to Critical. |

| Req. ID | Feature                      | Specification                                                                                                                                                                                                                                         | Alert Behaviour                                                                   |
|---------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| ESC-002 | Escrow Transaction Audit Log | Complete, chronological, immutable audit trail of all Escrow account transactions. Log is read-only. Exportable as PDF for official use.                                                                                                              | Any unauthorised balance discrepancy triggers a Level 3 alert immediately.        |
| ESC-003 | Monthly Escrow Health Report | Auto-generated monthly PDF per project: balance trend chart, compliance status, construction progress vs. Escrow adequacy, red-flag summary, and recommended actions (informational only). Automatically emailed to the assigned DLD project officer. | Report generated and delivered within 24 hours of month-end. Zero missed reports. |

## 9. Internal Platform — Developer Performance & Rating

### 9.1 Developer Scoring Engine

The Internal Platform must provide DLD staff with a comprehensive, data-driven developer rating system. The scoring model must be transparent, auditable, and configurable by authorised users. All scoring parameters and their weights must be adjustable from the platform's administration interface.

#### 9.1.1 Scoring Criteria

| Scoring Criterion               | Data Source                                                   | Default Weight | Calculation Method                                                                                                             |
|---------------------------------|---------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| On-Time Project Delivery Rate   | DLD project records + RERA delivery certificates              | 25%            | (Projects delivered on or before declared date ÷ Total completed projects) × 100. Weighted by project scale (unit count).      |
| Unit Sales Completion Rate      | DLD transaction registry                                      | 20%            | (Total units sold ÷ Total units registered across all projects) × 100.                                                         |
| Escrow Account Health Score     | RERA-certified Escrow Bank data feeds                         | 20%            | (Current Escrow balance ÷ Minimum required balance based on construction completion %) × 100.                                  |
| Regulatory Compliance Record    | RERA violations register + DLD audit records                  | 15%            | Score starts at 100. Deductions: Minor violation = -2, Major = -10, Critical = -25. Rolling 5-year window.                     |
| Financial Soundness Indicator   | Bank financial reports + credit assessments submitted to RERA | 10%            | Composite of liquidity ratios and debt-to-equity from most recent audited annual financial statements. Normalised 0–100 scale. |
| Historical Project Success Rate | DLD project completion records — 10-year history              | 10%            | (Successfully completed projects ÷ Total projects initiated in past 10 years) × 100.                                           |

#### 9.1.2 Developer Rating Interface Requirements

- Main Developer Leaderboard: sortable ranked list with composite score, individual criterion scores (mini radar chart per row), risk badge (High / Medium / Low), and trend arrow (score change vs. prior quarter).
- Developer Comparison: select up to 4 developers for side-by-side comparison — full radar chart overlay and criterion-by-criterion score table.
- Developer Profile Page: company overview, all current projects with individual risk ratings, Escrow balance per project, historical score trend chart, violation log, and total units delivered.
- Weight Configuration Panel: authorised administrator users can adjust the weight percentage assigned to each criterion. The system enforces that total weights always sum to 100%. All weight changes logged with timestamp and user identity.

- **Public Developer Scorecard (External Portal):** a simplified public-facing version showing completed projects count, on-time delivery percentage, RERA compliance rating, and total units delivered — supporting GRETI Governance sub-index improvement.

## 10. Non-Functional Requirements

### 10.1 Performance Standards

| Metric                          | Mandatory Target                                                                     | Measurement Method                                                       |
|---------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Homepage Initial Load Time      | Less than 3 seconds (P95, 10 Mbps connection)                                        | Industry-standard performance audit tool (vendor to specify in proposal) |
| API Response Time               | Less than 500 ms (P95) for all non-AI endpoints                                      | Application Performance Monitoring tool (vendor to specify)              |
| Interactive Map Filter Update   | Less than 2 seconds for any zone heatmap filter change                               | Automated UI performance tests                                           |
| AI Agent — Text Response        | Less than 8 seconds (P90) for factual data queries                                   | End-to-end latency monitoring with alerting                              |
| AI Agent — Chart Generation     | Less than 15 seconds (P90) for in-chat dashboard generation                          | End-to-end latency monitoring with alerting                              |
| Concurrent User Capacity        | 5,000 concurrent external users + 500 internal users without performance degradation | Load testing tools (vendor to specify)                                   |
| Platform Uptime (Availability)  | 99.9% per calendar month (maximum 8.7 hours downtime per year)                       | Uptime monitoring at 1-minute intervals + monthly SLA report             |
| Data Freshness — Dashboard KPIs | Refresh every 15 minutes                                                             | Data pipeline health monitoring + timestamp displayed on all KPIs        |
| Data Freshness — Transactions   | Lag from DLD source systems must not exceed 24 hours                                 | Automated pipeline monitoring with breach alerts                         |

### 10.2 Security & DESC Compliance

#### 10.2.1 ISR v3 Compliance Requirements

The platform must fully comply with the Dubai Electronic Security Center (DESC) Information Security Regulation (ISR) Version 3. The vendor must conduct a formal ISR applicability assessment and document compliance against all applicable control domains:

- Information Security Governance: defined information security policy, RACI matrix for security responsibilities, and annual security risk assessment.
- Asset Management: complete asset register, data classification per Dubai Government taxonomy, and asset owner assignment.
- Cryptography: industry-standard encryption for all data in transit and at rest. Secrets management solution proposed and implemented by the vendor.

- Access Control: RBAC enforced at the API layer. Principle of Least Privilege. MFA mandatory for all internal platform users. Session timeout: 30 minutes of inactivity for internal users.
- Incident Management: documented Security Incident Response Plan (SIRP). Notification to DLD CISO within 2 hours of detection. Notification to DESC within 24 hours for critical incidents.
- Third-Party Risk: all sub-contractors and third-party API providers must demonstrate security compliance with ISO 27001 or equivalent.

### 10.2.2 Security Testing Requirements

The following security testing activities are mandatory for the platform. The vendor must describe in their proposal how each activity will be executed, scheduled, and reported to DLD:

| Security Test Type         | Mandatory Requirements                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vulnerability Assessment   | Must be conducted using DESC-approved scanning tools before go-live of each phase, and thereafter on a regular basis as agreed with DLD and in compliance with DESC ISR v3 requirements. All Critical and High findings must be remediated within 30 days of discovery. Medium findings within 90 -120 days. Results must be made available to DLD on request.                           |
| Penetration Testing (VAPT) | A full penetration test must be completed by a DESC-approved independent third-party security assessor before the final platform go-live (Phase 3). Zero High or Critical findings may remain open at go-live. The VAPT report must be submitted to the DLD CISO. Thereafter, VAPT must be conducted in accordance with DESC ISR v3 requirements during the warranty and support period. |
| SOC / SIEM Monitoring      | The platform must be connected to a DESC-certified Security Operations Centre (SOC) providing continuous, real-time threat detection and incident alerting. The vendor must name the proposed SOC provider in their proposal and confirm their DESC certification status. Security incidents must be escalated to DLD in accordance with the incident classification in Section 15.2.    |
| OWASP Top 10 Compliance    | An automated OWASP Top 10 security scan must be integrated into the CI/CD deployment pipeline and executed with each deployment. Zero open Critical findings for any OWASP Top 10 category are permitted at go-live. Scan results must be available to DLD on request.                                                                                                                   |

### 10.3 Role-Based Access Control (RBAC) Matrix

| Role                | Permissions                                                                                                                | Authentication Requirements |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Public Visitor      | View all External Portal public data. No registration required for read-only access. Export requires CAPTCHA verification. | None / CAPTCHA for exports  |
| Registered Investor | All External Portal features + Watchlist + Personalised Alerts + Saved Analytics Views + Full Data Export.                 | Email + Password + MFA      |

| Role                 | Permissions                                                                                                                           | Authentication Requirements                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| DLD Staff — Viewer   | Read-only access to Internal Platform. Cannot modify anything.                                                                        | Centralised identity provider (vendor to propose) + MFA                                                             |
| DLD Staff — Operator | All Viewer permissions + CMS content management + Alert acknowledgement + Report generation.                                          | Centralised identity provider + MFA + Compliant device                                                              |
| DLD Supervisor       | All Operator permissions + User account management within division + EWRS threshold adjustment + Audit log review.                    | Centralised identity provider + MFA + Privileged Access approval                                                    |
| System Administrator | Full system access. AI model configuration. RBAC management. Scoring weight configuration. Infrastructure configuration via IaC only. | Centralised identity provider + MFA + Privileged Identity Management with Just-in-Time access and approval workflow |



## 11. Technology & Hosting Architecture — Vendor Proposal

### 11.1 Technology Stack — Vendor Proposal

DLD does not mandate any specific technology brand, framework, programming language, or cloud provider for application-layer components. Vendors are free to propose the technology stack they consider most appropriate to meet the functional, performance, security, and operational requirements described in this RFP. The chosen stack must be justified in the proposal on the grounds of fitness for purpose, security compliance, long-term maintainability, and total cost of ownership. DLD reserves the right to evaluate and challenge the proposed stack during the technical review stage. **RECOMMENDED DATA ARCHITECTURE APPROACH:** DLD has an existing data architecture based on the Microsoft Fabric ecosystem and strongly encourages vendors to align with and extend this environment. See Section 11.4 for DLD's data architecture preferences and guidance. Proposals demonstrating strong alignment with DLD's existing infrastructure will be evaluated more favourably.

The vendor's technology proposal must address all of the following architectural layers, specifying the proposed technology for each:

| Architectural Layer     | What the Vendor Must Specify & Justify                                                                                                                                                                                                                                                                                                                                                                                                            | Mandatory Standards                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend / Presentation | Proposed JavaScript framework or rendering approach. Justification for the choice. Approach to RTL/LTR bilingual layout. Component library and design system methodology.                                                                                                                                                                                                                                                                         | Lighthouse Performance Score greater than 85 (Desktop) and 75 (Mobile). WCAG 2.1 Level AA compliance. Full RTL support.                                                                           |
| Backend API Layer       | Proposed programming language, framework, and API style (REST, GraphQL, or hybrid). API Gateway solution. API documentation standard.                                                                                                                                                                                                                                                                                                             | P95 API response time under 500 ms. Rate limiting and DDoS protection on all public endpoints. OpenAPI 3.0 documentation mandatory.                                                               |
| AI / ML Services        | Multi-model AI orchestration layer design. Proposed primary and secondary AI models (any combination of managed API services, self-hosted open-source LLMs, or custom fine-tuned models). RAG architecture and retrieval engine. Model switching mechanism. Fallback design. UAE data residency confirmation for all active models.                                                                                                               | AI processing within UAE jurisdiction. RAG grounding mandatory. Hallucination rate below 5% on DLD data test set.                                                                                 |
| Database & Data Layer   | DLD strongly recommends that vendors build upon and integrate with its existing Microsoft Fabric ecosystem, including OneLake (Silver and Gold layers), Data Factory pipelines, existing semantic models, and the reporting layer, as this maximises DLD's existing investments and reduces implementation risk. Vendors proposing a different approach must provide a clear justification and demonstrate equivalent or superior capability. See | All data must flow through OneLake Silver and Gold layers. OLTP source data: under 10 ms read for single-row queries. OneLake/Fabric OLAP: under 2 seconds for 100M row aggregations. No parallel |

| Architectural Layer             | What the Vendor Must Specify & Justify                                                                                                                         | Mandatory Standards                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                                 | Section 11.4 for the full data architecture guidance. For the External Public Portal visualisation layer, see Section 11.4.2.                                  | or duplicate data stores permitted outside the Fabric ecosystem without written DLD approval.                                  |
| GIS / Mapping                   | Proposed mapping library and GIS data source. How official Dubai Municipality zone boundaries will be used and kept current.                                   | Map loads within 5 seconds. Heatmap layer switches within 2 seconds.                                                           |
| Content Management System (CMS) | Proposed CMS product and whether self-hosted or SaaS. DESC-CSP compliance approach if SaaS. API-first architecture confirmation.                               | Content changes live within 30 seconds of publish. Admin interface usable by non-technical staff.                              |
| Authentication & Identity       | Proposed identity provider and authentication protocol. Integration approach with DLD's existing identity management. MFA mechanism.                           | OAuth 2.0 + OIDC standard. MFA mandatory for all internal users. All authentication events in immutable audit log.             |
| Infrastructure & DevOps         | Proposed containerisation and orchestration approach. CI/CD pipeline tools. Infrastructure as Code (IaC) methodology. Monitoring, logging, and alerting stack. | IaC mandatory — 100% of infrastructure defined as code. Zero-downtime deployments. CI/CD includes automated security scanning. |

## 11.2 Hosting Model — Vendor Options

DLD does not specify a hosting provider or model. The vendor must propose a hosting solution meeting the security, data residency, performance, and regulatory requirements of this RFP. Vendors may propose one or more of the following hosting models, each with a separately itemised cost structure.

| Hosting Model                    | Description                                                                                                                                                                                       | Mandatory Requirements for This Option                                                                                                          |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Public Cloud (UAE Region)        | Deployment on a public cloud platform in a UAE data centre region. Vendor must name the specific provider and confirm DESC-CSP certification status.                                              | DESC-CSP certification confirmed. All data resides in the UAE. Multi-availability-zone deployment. RTO less than 4 hours, RPO less than 1 hour. |
| Private Cloud / Colocation (UAE) | Deployment in a private cloud or colocation data centre within the UAE. Vendor must name the data centre, confirm its tier rating (minimum Tier III), and describe the hardware management model. | Data centre physically in the UAE. ISO 27001 certified facility. Disaster recovery site in a separate UAE data centre.                          |

| Hosting Model                 | Description                                                                                                                                                                                                 | Mandatory Requirements for This Option                                                                                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| On-Premises / DLD Data Centre | Deployment on infrastructure hosted within DLD's own data centre or UAE government data centre. Vendor supplies hardware specifications.                                                                    | Vendor must supply a detailed Hardware Bill of Materials (BOM) with procurement costs priced separately from implementation.                            |
| Hybrid Model                  | A combination of the above — e.g., public cloud for the External Portal and on-premises for the Internal Platform. Vendor must clearly describe the data partitioning, connectivity, and security boundary. | All inter-system data transfers encrypted. Latency between hybrid components must not breach performance SLAs. Full DESC compliance for all components. |

### 11.3 Mandatory Hosting Requirements (All Models)

- **Data Residency:** all data — production, backup, and disaster recovery — must remain within the geographic boundaries of the United Arab Emirates at all times.
- **DESC-CSP Compliance:** the hosting infrastructure must be provided by, or hosted within a facility certified by, DESC as a Cloud Service Provider (CSP), or must achieve DESC compliance through an alternative pathway approved in writing by DESC.
- **High Availability:** no single point of failure for any critical component. Target availability: 99.9% per calendar month.
- **Disaster Recovery:** a documented Disaster Recovery Plan (DRP) must be provided. RTO: less than 4 hours. RPO: less than 1 hour.
- **Encrypted Backups:** daily automated encrypted backups with a minimum 30-day retention. Backup restoration tested quarterly with results reported to DLD.
- **Infrastructure as Code:** all infrastructure defined and deployable as code. DLD must be provided with all IaC scripts upon project completion.

### 11.4 Mandatory DLD Data Architecture & Infrastructure Requirements

#### **STRONGLY RECOMMENDED: Utilisation of DLD's Existing Data Architecture & Infrastructure**

DLD strongly recommends that vendors fully leverage and build upon the existing data architecture and infrastructure currently implemented by the Dubai Land Department (DLD). Proposals that align with DLD's existing ecosystem will be evaluated more favourably, as they maximise the value of current investments, reduce integration risk, and accelerate delivery. Vendors proposing an alternative approach are not excluded, but must provide a detailed technical justification demonstrating why the existing ecosystem cannot meet the requirement and how their proposed alternative delivers equivalent or superior outcomes. The recommended components are as follows:

- **On-Premises Environment:** It is recommended that the solution be designed to operate within and integrate with DLD's current on-premises infrastructure where applicable. Vendors are encouraged to leverage existing on-premises components rather than introduce external hosting for data layer components. Where a vendor proposes a different hosting arrangement for specific data components, a clear justification and description of how this aligns with DESC compliance and DLD data governance standards should be provided.

- **Microsoft Fabric Ecosystem — OneLake (Lakehouse Architecture):** It is strongly recommended that vendors utilise DLD's existing OneLake environment, structured with established Silver and Gold data layers. Aligning data ingestion, transformation, and serving with this Lakehouse architecture avoids duplication and preserves data governance consistency. Vendors proposing an alternative data storage approach must justify why OneLake cannot fulfil the requirement.
- **Microsoft Fabric — Data Factory Pipelines:** Vendors are strongly encouraged to build upon DLD's existing Data Factory pipeline infrastructure within the Microsoft Fabric ecosystem for all data ingestion and transformation work. This approach preserves operational continuity and avoids introducing a separate ETL or orchestration layer. Where a vendor believes an additional or alternative pipeline component is technically necessary, this must be clearly identified and justified in the proposal.
- **Semantic Models & Reporting Layer:** Vendors are encouraged to integrate with and extend DLD's existing semantic models and reporting layer within the Fabric ecosystem. Where additional semantic models or computed measures are required, building these as extensions to the current model structure is preferred, maintaining consistency with existing data definitions, naming conventions, and governance standards. Vendors proposing a different semantic modelling approach should explain how data consistency and governance will be maintained.
- **Existing Data Exchange & Integration Mechanisms:** Vendors are encouraged to leverage DLD's current data exchange and integration mechanisms for connecting to source systems, avoiding unnecessary duplication of integration infrastructure. Where a vendor identifies a gap in the existing integration layer that cannot be addressed within the current mechanisms, this should be clearly documented in the proposal with a proposed approach for DLD's review.

**General Architecture Guidance:** DLD's preference is for all proposed solutions to integrate seamlessly within the current platform, ensuring consistency with existing data models, governance standards, security controls, and operational processes, while maximising the value of current investments and avoiding fragmentation or redundancy across systems. Proposals that align with this preference will be scored more favourably in the technical evaluation. Vendors who believe a specific component is better served by an alternative approach are welcome to propose it, provided they include a clear technical rationale and demonstrate how data consistency, security, and governance will be preserved.

#### **11.4.1 Recommended Vendor Deliverables — Data Architecture Alignment**

- An Architecture Integration Map — recommended to be submitted as part of the proposal — showing how each proposed component connects to and leverages the existing Microsoft Fabric environment (OneLake layers, Data Factory pipelines, semantic models, and integration mechanisms). Where the vendor proposes components outside the Fabric ecosystem, these should be clearly identified and justified.
- Where a vendor identifies requirements they consider better addressed outside the existing ecosystem, a brief justification register is recommended, identifying each component with a technical rationale and a proposed compliant alternative. This transparency is welcomed and will be evaluated constructively.
- A Data Architecture Familiarisation Plan describing how the vendor proposes to assess and document DLD's current Microsoft Fabric configuration, OneLake schema, pipeline topology, and semantic model structure during the discovery phase in Section 12.1, prior to development commencing. This ensures proposals are grounded in DLD's actual environment rather than assumptions.

#### **11.4.2 Visualisation Layer — External Public Portal**

While DLD's internal reporting layer within the Microsoft Fabric ecosystem may utilise Power BI for internal dashboards and management reporting, Power BI is recognised as potentially unsuitable as the primary visualisation engine for the External Public Portal due to the following considerations: anonymous public access and licensing model constraints, cost scalability under high concurrent external user volumes, embedding complexity for fully custom investor-facing interfaces, and public portal UX/UI customisation limitations.

Accordingly, for the External Public Portal (investor-facing and public-access components), vendors are explicitly permitted and encouraged to propose an alternative, purpose-built visualisation and analytics layer in place of or supplementary to Power BI, subject to the following conditions:

- The proposed external visualisation solution is strongly encouraged to consume data directly from DLD's OneLake Gold layer or from API endpoints served from the existing Fabric semantic models, maintaining the Gold layer as the preferred single source of truth and avoiding unnecessary data duplication.
- The vendor must provide a comparative analysis in their proposal evaluating Power BI embedding against their proposed alternative, addressing: total licensing cost at projected public user volumes, anonymous access capability, UI/UX customisation depth, performance at scale (concurrent users), and long-term maintainability.
- Acceptable alternative visualisation technologies include but are not limited to: custom-built web charting libraries (such as Apache ECharts, D3.js, Highcharts, or Recharts), embedded analytics platforms designed for public portals (such as Metabase, Redash, Grafana, or equivalent), or a fully custom-developed analytics front-end consuming the Fabric-served data APIs.
- If the vendor proposes Power BI for any element of the External Public Portal, they must include a full licensing cost model in Section 14, demonstrating cost-effectiveness at a minimum of 10,000 concurrent anonymous external users, together with a confirmed description of how anonymous public access will be technically achieved within the Power BI licensing framework.
- Power BI or Fabric-native reporting tools remain fully applicable and preferred for all Internal Platform components (EWRS dashboards, Developer Performance dashboards, and internal management reporting), where the licensing model is appropriate for named internal users.

## 12. DLD Data Understanding & Analytics Assessment

The accuracy and credibility of all data published on the IRETP platform is the primary determinant of the platform's value to international investors and its impact on GRETI rankings. The vendor must treat data accuracy as a first-class deliverable throughout both development and the post-go-live support period. In line with DLD's strong preference for vendors to leverage the existing Microsoft Fabric ecosystem (Section 11.4), the vendor's data familiarisation activities should include an assessment of the current OneLake Lakehouse architecture (Silver and Gold layers), Data Factory pipeline configurations, semantic model definitions, and existing data exchange and integration mechanisms. This will enable the vendor to propose the most effective integration approach and identify any gaps or areas where alternative solutions may be required.

### 12.1 DLD Data Familiarisation — Mandatory Discovery

Before any data pipeline development begins, the vendor must complete a formal DLD Data Familiarisation exercise. DLD will assign a dedicated data liaison officer and a Microsoft Fabric data engineer to support this exercise. In addition to the source system inventory activities below, the vendor must conduct a full assessment of DLD's existing Microsoft Fabric environment, including the OneLake Lakehouse schema (Silver and Gold layer structure), active Data Factory pipeline configurations, published semantic model definitions, and all current data exchange and integration mechanisms. The following outputs are mandatory:

| Familiarisation Area             | Required Activities                                                                                                                                                                                                                                                                                                  | Output Document                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Source System Inventory          | Catalogue all DLD source systems feeding the IRETP platform: transaction registry, project database, Ejari rental system, RERA regulatory records, Escrow bank feeds. Document for each: system name, data format, update frequency, API or extract availability, and responsible DLD department.                    | DLD Data Source Inventory — to be approved by DLD Data Liaison Officer |
| Data Schema & Field Mapping      | Review and document the data schema of each source system: field names, data types, allowable values, null/empty conventions, and any coding or classification schemes used by DLD (transaction type codes, property category codes, zone identifiers, developer licence number formats).                            | Detailed Field Mapping Document — to be approved by DLD IT             |
| Data Quality Baseline Assessment | Profile the data in each source system to establish a quality baseline covering: completeness rate per field, uniqueness (duplicate record identification), consistency (cross-system alignment), and known data quality issues. Document all discovered quality issues and agree on the handling approach with DLD. | Data Quality Baseline Report — to be submitted to DLD for approval     |

| Familiarisation Area               | Required Activities                                                                                                                                                                                                                                                             | Output Document                                                              |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Business Logic & Calculation Rules | Validate with DLD the business logic used to calculate all derived metrics published on the platform: price per sqft formula, rental yield, Escrow adequacy ratio, developer score, and risk indicator trigger logic. Any ambiguity must be resolved before development begins. | Calculation Rules Validation Document — to be approved by DLD Analytics team |
| Historical Data Assessment         | Assess the completeness and consistency of DLD's historical transaction data for the 5-year period that will be published. Document all gaps, anomalies, and quality issues. Agree a data remediation plan with DLD before the historical data is loaded.                       | Historical Data Assessment Report — to be submitted to DLD for approval      |

## 12.2 Pre-Publication Analytics Assessment

Before any data is made live on the External Portal, the vendor must complete a Pre-Publication Analytics Assessment. This assessment is a mandatory gate — no public data publication is permitted until the assessment is completed and signed off by the DLD data liaison officer. The assessment must validate:

- End-to-End Data Reconciliation: platform figures match the authoritative DLD source system records for a minimum sample of 500 transactions drawn randomly across all property types, zones, and transaction types. Accuracy rate must be 99.5% or above.
- Derived Metric Validation: every derived metric (price per sqft averages, rental yield %, zone-level aggregates, YoY change figures, Escrow adequacy ratios, developer scores) must be independently validated from raw data. Any discrepancy must be resolved before go-live.
- Edge Case & Outlier Review: all statistical outliers must be identified, flagged to DLD, and handled (included, excluded, or annotated) by agreement before go-live.
- Zone Boundary Verification: zone-level data aggregations use the correct official zone boundary definitions. No transactions double-counted or misassigned.
- Trend Direction Validation: all trend indicators (up/stable/down arrows, YoY % changes) verified against raw data for correct direction and magnitude.

## 12.3 Data Accuracy Obligations During Warranty & Support Period

After go-live, ongoing data monitoring, reconciliation, and periodic reporting are the responsibility of DLD. The vendor's data accuracy obligations during the 12-month warranty and technical support period are limited to the following:

- Data Error Investigation & Correction: if DLD or any platform user reports a suspected data inaccuracy, the vendor must investigate and provide a written response within 2 business hours, with a correction deployed within 24 hours if the error is confirmed to be a defect in the platform's data pipeline or transformation logic.
- Post-DLD-Update Validation: whenever DLD modifies any source system data structure (new fields, changed formats, updated zone classifications, new transaction type codes) and

- notifies the vendor, the vendor must update the platform's data pipeline, field mappings, and analytics calculations accordingly, and confirm accuracy of the updated data before it is published, within 10 business days of receiving DLD's change specification.
- Pipeline Failure Resolution: if the vendor's monitoring detects a data pipeline failure causing data to go stale beyond the freshness thresholds in Section 10.1, the vendor must resolve the failure and restore data freshness within the P1/P2 SLA timelines in Section 15.2.



### 13. Phased Delivery Plan

**MANDATORY TIMELINE CONSTRAINT:** The total project duration — covering all four delivery phases — must not exceed 3 calendar months (120 days) from the date of contract signing. Vendors must demonstrate in their proposal how they will deliver all four phases within this constraint. Proposals that cannot credibly justify delivery within 120 days will be evaluated accordingly. **GO-LIVE GATE:** No phase may go live or be made publicly accessible until all mandatory testing and acceptance criteria for that phase have been fully passed and signed off by the DLD Project Sponsor. The completion of testing is a prerequisite for go-live, not a parallel activity. The vendor bears full responsibility for scheduling testing with sufficient time to resolve all defects before the 120-day deadline.

Each phase has defined deliverables that constitute the sole acceptance gate for that phase. Within the 120-day constraint, vendors must allocate time in their proposed Gantt chart for: development, integration, testing (functional, performance, security, accessibility, and data accuracy), defect resolution, and DLD sign-off. Vendors are not required to itemise cost per individual task within a phase — cost is presented per phase as described in Section 14.

#### Phase 1 — External Public Portal

Phase 1 delivers the complete, publicly accessible External Portal in English and Arabic on production infrastructure. This is the highest-priority phase and must be completed and go-live ready — with all testing passed — within the vendor's proposed Phase 1 timeline, which must fall within the overall 120-day constraint. No data may be published publicly until the Pre-Publication Analytics Assessment has been completed and signed off by DLD.

Phase 1 must include the following deliverables:

12. DLD Data Familiarisation — all five output documents completed and approved by DLD (source inventory, field mapping, data quality baseline, calculation rules validation, historical data assessment).
13. Pre-Publication Analytics Assessment completed and signed off by DLD data liaison officer prior to go-live.
14. External Portal UI in English and Arabic with full RTL support, responsive design (desktop, tablet, mobile), and WCAG 2.1 Level AA compliance.
15. Interactive GIS Map with three switchable zone heatmap layers (transaction activity, average price per sqft, rental yield) and project pins with detail panels.
16. Transactions Page with full 5-year historical data, multi-dimensional filters, and export in Excel, CSV, and PDF formats.
17. Price Index Page with 10-year trend chart and up to 5-zone comparison.
18. Projects Database Page with all project statuses, filtering, and zone-linked map.
19. Rental Index Page with yield calculator and zone comparison.
20. AI Agent (External) with RAG layer — full conversational capability in English and Arabic, in-chat chat generation, and download in all four formats.
21. Interactive Analytics Engine (Slice & Dice) — all 9 visualisation types, saved views, personal dashboards, and shareable links.

22. Headless CMS deployed and DLD non-technical staff trained.
23. Investor registration, Watchlist, and full alert system (all 6 alert types) with email and SMS delivery.
24. Open Data API with developer portal, API key management, and OpenAPI 3.0 documentation — 5 standard endpoints operational.
25. Production infrastructure live on the vendor's proposed hosting solution, fully DESC-compliant.
26. Load test demonstrating acceptable performance under the concurrent user loads specified in Section 10.1.

## Phase 2 — Early Warning Risk System (EWRs)

Phase 2 delivers the Internal Platform's Early Warning Risk System and Escrow Monitoring capabilities to DLD staff and leadership.

Phase 2 must include the following deliverables:

27. EWRs Dashboard with Dubai-wide risk heatmap on GIS map, risk KPI summary bar, alert inbox, and risk trend charts.
28. All 10 risk indicators operational with configurable thresholds adjustable from the administration panel.
29. Multi-level alert routing delivering alerts to all 4 leadership levels through correct channels (email, SMS, encrypted email, auto-generated PDF briefing).
30. Playbook integration linking each alert type to a pre-defined SOP checklist.
31. Escrow Account Monitoring Dashboard (per-project) with live balance data, adequacy ratio, immutable transaction audit log, and automated monthly PDF report.
32. Full RBAC for all 6 internal user roles with identity provider integration and MFA enforcement.
33. Internal AI Agent with full conversational data capability and in-chat dashboard generation with download functionality.
34. Phase 2 load test and UAT sign-off.

## Phase 3 — Developer Performance & Rating

Phase 3 delivers the Internal Developer Performance and Rating platform, including the public-facing Developer Scorecard.

Phase 3 must include the following deliverables:

35. Developer Rating Engine operational with all 6 scoring criteria connected to live DLD and RERA data feeds.
36. Developer Leaderboard with composite scores, radar chart per developer, risk badges, and trend arrows.
37. Developer Comparison view supporting up to 4 developers side-by-side.

38. Developer Profile Page with full project portfolio, Escrow balance per project, historical score trend, and violation log.
39. Weight Configuration Panel allowing authorised administrators to adjust scoring weights, with changes logged with timestamp and user identity.
40. Geographic map view of all developer projects, colour-coded by risk rating.
41. Public Developer Scorecard on the External Portal showing simplified metrics for international investors.
42. DESC-authorised Penetration Test (VAPT) completed, all High and Critical findings remediated, and VAPT report signed by DESC-approved assessor.
43. ISO 27001 gap assessment completed and certification roadmap presented and approved by DLD.
44. Phase 3 UAT sign-off.

## Phase 4 — Extended Languages for the External Portal

Phase 4 delivers full multilingual support for the External Portal beyond English and Arabic, expanding accessibility for global investors.

Phase 4 must include the following deliverables:

45. Full UI translation and static content localisation for all six extended languages: Simplified Chinese, Russian, Urdu, French, Hindi, and German.
46. RTL layout correctly applied for Urdu across all portal pages and components. No untranslated strings visible in any language.
47. AI Agent basic query handling in all six extended languages with graceful English fallback for complex queries.
48. Arabic translation of official zone names, project names, and developer names validated by DLD official records (also applies to new language additions).
49. All six language versions tested for completeness, layout correctness, and data label accuracy.
50. ESG / Sustainability Module: LEED and Estidama Pearl certification data displayed for applicable projects on the External Portal and on the GIS map as an additional heatmap layer.
51. International Market Benchmarking module: Dubai key metrics compared against 5 global peer cities (London, Singapore, New York, Paris, Hong Kong), updated quarterly.
52. PDF Investment Profile Generator: zone-level and project-level PDF generation with DLD branding, charts, KPIs, and embedded maps, generated within 10 seconds.
53. Complete technical documentation package: system architecture documentation, API reference (all endpoints), operational runbooks, disaster recovery playbook, and data dictionary.
54. DLD staff training programme — 8 structured sessions covering all platform roles, CMS management, EWRS operation, and AI Agent best practices.
55. Full final UAT across all four phases. DLD Project Sponsor signs final UAT acceptance form.

## 14. Cost Structure

The cost structure is separated into four distinct categories to provide DLD with full visibility over the financial commitment associated with this project. Vendors must complete all four cost tables. Bundling of items across categories is not permitted. All prices must be stated in UAE Dirhams (AED), exclusive of VAT.

### 14.1 Infrastructure & Hosting Setup Costs (One-Time CAPEX)

This table covers the cost of procuring, provisioning, configuring, and certifying the infrastructure required to host and operate the platform. These are one-time capital costs and are separate from the application development costs in Section 14.2. Where the vendor proposes multiple hosting options, a separate column must be completed for each option.

| #    | Infrastructure Setup Item                                                                                                                                 | Option A (AED) | Option B (AED) |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| I.1  | Compute infrastructure setup (servers, virtual machines, containers, orchestration cluster) — production environment. Vendor to specify configuration.    | _____          | _____          |
| I.2  | Compute infrastructure setup — staging and development environments.                                                                                      | _____          | _____          |
| I.3  | Database infrastructure setup: OLTP database server(s), OLAP/analytics data warehouse, caching layer. Vendor to specify configuration.                    | _____          | _____          |
| I.4  | Storage infrastructure: object/file storage, backup storage. Vendor to specify capacity.                                                                  | _____          | _____          |
| I.5  | Networking infrastructure: load balancers, firewalls, Web Application Firewall (WAF), DDoS protection, VPN, API Gateway hardware or configuration.        | _____          | _____          |
| I.6  | Content Delivery Network (CDN) setup and configuration.                                                                                                   | _____          | _____          |
| I.7  | SSL/TLS certificates — initial procurement and installation.                                                                                              | _____          | _____          |
| I.8  | Disaster Recovery site setup (secondary environment in a separate UAE data centre or zone).                                                               | _____          | _____          |
| I.9  | Security infrastructure: SIEM/SOC integration setup, vulnerability scanning tool deployment, secrets management system, encryption key management system. | _____          | _____          |
| I.10 | Identity and access management (IAM) infrastructure setup and integration with DLD's identity systems.                                                    | _____          | _____          |
| I.11 | GIS/Mapping service setup and initial configuration.                                                                                                      | _____          | _____          |
| I.12 | AI/LLM inference infrastructure setup (if self-hosted or private cloud model — GPU servers, model serving infrastructure, etc.).                          | _____          | _____          |

| #                                                       | Infrastructure Setup Item                                                                                                                        | Option A (AED) | Option B (AED) |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| I.13                                                    | DESC-CSP certification process costs (certification fees, assessor fees, documentation).                                                         | _____          | _____          |
| I.14                                                    | Initial security penetration test / VAPT (pre-go-live) — assessor fees.                                                                          | _____          | _____          |
| I.15                                                    | Hardware procurement (on-premises option only): servers, storage, networking hardware. Itemised Bill of Materials to be attached as an appendix. | _____          | N/A            |
| <b>TOTAL INFRASTRUCTURE SETUP COST (One-Time CAPEX)</b> |                                                                                                                                                  |                |                |
|                                                         | TOTAL:                                                                                                                                           | _____          | _____          |

## 14.2 Software Licences & Subscriptions — Setup / First Year (CAPEX / Upfront)

This table covers the cost of software licences, SaaS platform subscriptions, and API service enrolments required to build and operate the platform. Where a licence has a recurring annual component, the first-year cost should be stated here; the recurring component must also appear in Section 14.4. Where multiple hosting options are proposed, provide a separate column.

| #   | Licence / Subscription Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Option A (AED) | Option B (AED) |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| L.1 | CMS platform licence or subscription (first year) — vendor to specify the proposed CMS product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | _____          | _____          |
| L.2 | AI / LLM service licence or API subscription (first year) — vendor to specify the proposed service and pricing model (per token, per request, fixed tier, etc.).                                                                                                                                                                                                                                                                                                                                                                                                                | _____          | _____          |
| L.3 | GIS / Mapping API licence or subscription (first year) — vendor to specify the proposed service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | _____          | _____          |
| L.4 | Email delivery service licence or subscription (first year) — vendor to specify the proposed service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | _____          | _____          |
| L.5 | SMS gateway service licence or subscription (first year) — vendor to specify the proposed service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | _____          | _____          |
| L.6 | External Public Portal visualisation layer licence or subscription (first year) — vendor to specify the proposed technology (e.g. Power BI Embedded with full licensing cost model at 10,000+ concurrent users, or alternative charting/analytics platform as per Section 11.4.2). If Power BI is not proposed for the External Portal, vendor must specify the alternative and confirm it is zero-licence-cost or provide the licence cost. If no additional licence is required (open-source or custom-built solution), enter AED 0 and state “No external licence required”. | _____          | _____          |
| L.6 | Monitoring, logging, and observability platform licence (first year) — vendor to specify.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | _____          | _____          |

| #                                                         | Licence / Subscription Item                                                                                                 | Option A (AED) | Option B (AED) |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| L.7                                                       | Security scanning / vulnerability management tool licence (first year) — vendor to specify (must be DESC-approved).         | _____          | _____          |
| L.8                                                       | Identity / SSO platform licence (first year) — if using a commercial identity provider.                                     | _____          | _____          |
| L.9                                                       | Third-party data feed subscriptions (currency rates, ESG certification data, international benchmarking data) — first year. | _____          | _____          |
| L.10                                                      | Any other software licences or subscriptions required — vendor to itemise.                                                  | _____          | _____          |
| <b>TOTAL LICENCE &amp; SUBSCRIPTION COST (First Year)</b> |                                                                                                                             |                |                |
|                                                           | TOTAL:                                                                                                                      | _____          | _____          |

### 14.3 Application Development & Implementation Costs (CAPEX) — By Phase

This table covers the cost of designing, developing, integrating, testing, and delivering the platform across four phases. Infrastructure and licence costs must not be included here — they are captured in Sections 14.1 and 14.2. The vendor must provide a total price per phase and a total project implementation price.

| Phase / Deliverable                                                               | Implementation Cost (AED) |
|-----------------------------------------------------------------------------------|---------------------------|
| <b>PHASE 1 — External Public Portal</b>                                           |                           |
| DLD Data Familiarisation & Pre-Publication Analytics Assessment (all outputs)     | _____                     |
| UX/UI Design and Design System                                                    | _____                     |
| External Portal Frontend Development (all pages, English + Arabic, RTL)           | _____                     |
| Backend APIs, Data Integration, and Data Pipeline Development                     | _____                     |
| AI Agent v1 — External Platform (LLM integration, RAG, in-chat charts, downloads) | _____                     |
| Interactive Analytics Engine (Slice & Dice) — all features                        | _____                     |
| CMS Setup and DLD Staff Training                                                  | _____                     |
| Investor Registration, Watchlist, and Alert Engine                                | _____                     |
| Data Export Engine (Excel, CSV, PDF)                                              | _____                     |
| Open Data API and Developer Portal                                                | _____                     |
| QA, Testing, Data Accuracy Verification, and Phase 1 UAT                          | _____                     |
| <b>PHASE 1 TOTAL</b>                                                              |                           |

| Phase / Deliverable                                                              | Implementation Cost (AED) |
|----------------------------------------------------------------------------------|---------------------------|
|                                                                                  | _____                     |
| <b>PHASE 2 — Early Warning Risk System (EWRs)</b>                                |                           |
| EWRs Dashboard and all 10 risk indicators with configurable thresholds           | _____                     |
| Multi-level alert routing (all 4 levels and all delivery channels)               | _____                     |
| Playbook integration and alert audit trail                                       | _____                     |
| Escrow Monitoring Dashboard, audit log, and monthly automated report             | _____                     |
| Full RBAC for all 6 internal roles and identity provider integration with MFA    | _____                     |
| Internal AI Agent with in-chat dashboard generation and download                 | _____                     |
| QA, Testing, and Phase 2 UAT                                                     | _____                     |
| <b>PHASE 2 TOTAL</b>                                                             |                           |
|                                                                                  | _____                     |
| <b>PHASE 3 — Developer Performance &amp; Rating</b>                              |                           |
| Developer Rating Engine with all 6 scoring criteria connected to live data feeds | _____                     |
| Developer Leaderboard, Comparison view, and Profile pages                        | _____                     |
| Weight Configuration Panel and geographic map view                               | _____                     |
| Public Developer Scorecard on External Portal                                    | _____                     |
| DESC-authorised VAPT and remediation of all High/Critical findings               | _____                     |
| ISO 27001 Gap Assessment and certification roadmap                               | _____                     |
| QA, Testing, and Phase 3 UAT                                                     | _____                     |
| <b>PHASE 3 TOTAL</b>                                                             |                           |
|                                                                                  | _____                     |
| <b>PHASE 4 — Extended Languages &amp; Supplementary Features</b>                 |                           |
| Full UI translation and static content localisation — 6 extended languages       | _____                     |
| AI Agent basic query handling — 6 extended languages                             | _____                     |
| ESG / Sustainability Module (LEED/Estidama data layer and heatmap)               | _____                     |
| International Market Benchmarking module — Dubai vs. 5 global cities             | _____                     |
| PDF Investment Profile Generator (zone and project level)                        | _____                     |

| Phase / Deliverable                                                         | Implementation Cost (AED) |
|-----------------------------------------------------------------------------|---------------------------|
| Complete Technical Documentation Package                                    | _____                     |
| DLD Staff Training Programme — 8 structured sessions                        | _____                     |
| Final UAT across all four phases and go-live support (hypercare)            | _____                     |
| <b>PHASE 4 TOTAL</b>                                                        |                           |
|                                                                             | _____                     |
| <b>TOTAL APPLICATION DEVELOPMENT &amp; IMPLEMENTATION COST (All Phases)</b> |                           |
|                                                                             | _____                     |

#### 14.4 Annual Recurring Costs (OPEX) — Per Hosting Option

This table covers all recurring annual costs for operating and maintaining the platform after go-live. Infrastructure costs, licence renewals, support costs, and monitoring costs must all be included. Vendors must provide a separate column for each hosting option proposed. The 12-month post-go-live support cost from Section 15 must also appear here for a complete annual picture.

| #                                       | Annual Recurring Cost Item                                                                                                                                          | Option A (AED/year) | Option B (AED/year) |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| <b>Infrastructure &amp; Hosting</b>     |                                                                                                                                                                     |                     |                     |
| R.1                                     | Annual hosting / infrastructure costs (compute, storage, networking, CDN, databases) — based on actual projected load. Vendor must provide calculation methodology. | _____               | _____               |
| R.2                                     | Disaster recovery site annual operating cost.                                                                                                                       | _____               | _____               |
| R.3                                     | Hardware maintenance and refresh allowance (on-premises option only).                                                                                               | _____               | N/A                 |
| <b>Software Licences &amp; Services</b> |                                                                                                                                                                     |                     |                     |
| R.4                                     | CMS platform annual licence or subscription renewal.                                                                                                                | _____               | _____               |
| R.5                                     | AI / LLM service annual cost — based on projected 5,000 daily active users and average query volume. Vendor must show the calculation.                              | _____               | _____               |
| R.6                                     | GIS / Mapping API annual costs — based on projected annual map loads and API calls.                                                                                 | _____               | _____               |
| R.7                                     | Email delivery and SMS gateway annual costs — based on estimated monthly alert volumes.                                                                             | _____               | _____               |
| R.8                                     | Monitoring, logging, and observability platform annual licence renewal.                                                                                             | _____               | _____               |
| R.9                                     | Identity / SSO platform annual licence renewal.                                                                                                                     | _____               | _____               |



| #                                                         | Annual Recurring Cost Item                                                                                         | Option A<br>(AED/year) | Option B<br>(AED/year) |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|
| R.10                                                      | Third-party data feed subscriptions annual renewal (currency rates, ESG data, benchmarking data).                  | _____                  | _____                  |
| R.11                                                      | SSL/TLS certificate annual renewal.                                                                                | _____                  | _____                  |
| <b>Security &amp; Compliance</b>                          |                                                                                                                    |                        |                        |
| R.12                                                      | Annual Penetration Test + Quarterly Vulnerability Assessment fees (DESC-approved assessor).                        | _____                  | _____                  |
| R.13                                                      | 24/7 SOC/SIEM Monitoring Service annual cost (DESC-certified SOC provider — vendor to name the proposed provider). | _____                  | _____                  |
| R.14                                                      | DESC-CSP Annual Surveillance Audit fees.                                                                           | _____                  | _____                  |
| R.15                                                      | ISO 27001 Annual Surveillance Audit fees.                                                                          | _____                  | _____                  |
| <b>Warranty &amp; Technical Support (from Section 15)</b> |                                                                                                                    |                        |                        |
| R.16                                                      | 12-month post-go-live support and maintenance annual cost (as detailed in Section 15.4).                           | _____                  | _____                  |
| <b>TOTAL ANNUAL RECURRING COST (OPEX)</b>                 |                                                                                                                    |                        |                        |
|                                                           | TOTAL:                                                                                                             | _____                  | _____                  |

## 15. Warranty & Technical Support — 12 Months Post Go-Live

Following the final UAT sign-off and go-live of all four phases, the vendor is contractually obligated to provide a 12-month warranty and technical support period. This obligation is included within the scope of this RFP and must be fully priced within the vendor's proposal. It is not optional and is not a separate procurement.

The ongoing monitoring of data quality, periodic data reconciliation, and the generation of operational and analytics reports are the responsibility of DLD. The vendor's obligations during the warranty and support period are limited to the categories set out below.

### 15.1 Warranty & Support Scope

| Support Category                | Scope Detail                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Defect Resolution (Warranty)    | All defects identified after go-live — whether reported by DLD staff, platform users, or detected by the vendor's monitoring tools — will be investigated and resolved by the vendor at no additional cost. A defect is defined as any behaviour that deviates from the accepted UAT test cases or the specifications in this RFP. Defect resolution is classified and responded to according to the SLA in Section 15.2.  |
| Data Pipeline Defect Correction | If DLD identifies or reports a data accuracy issue caused by a defect in the vendor's data pipeline, transformation logic, or calculation implementation, the vendor must investigate and respond within 2 business hours and deploy a correction within 24 hours if the error is confirmed. Periodic data reconciliation and ongoing accuracy monitoring are DLD's responsibility.                                        |
| Security Patch Management       | The vendor must monitor all platform software components, libraries, and infrastructure for published security vulnerabilities and apply patches proactively within: Critical (CVSS 9.0 and above) within 24 hours; High (CVSS 7.0–8.9) within 72 hours; all others within 14 days. Patch activities must be documented and available to DLD on request.                                                                   |
| Performance Defect Resolution   | If the vendor's monitoring detects, or DLD reports, a sustained breach of the performance SLAs in Section 10.1 (breach lasting 3 or more consecutive days), the vendor must investigate root cause and implement a fix within 48 hours at no additional cost.                                                                                                                                                              |
| DLD Data Structure Changes      | If DLD modifies any source system data structure (new fields, changed formats, updated zone classifications, new transaction type codes) and formally notifies the vendor, the vendor must update the platform's data pipeline, field mappings, and analytics calculations within 10 business days of receiving DLD's written change specification. Accuracy of the updated data must be confirmed before it is published. |
| Minor Changes Pool              | A pool of 120 person-hours of minor change capacity is included within the warranty and support contract at no additional cost. This covers adjustments to existing functionality (threshold adjustments, dashboard layout changes, new filter options, CMS template updates, alert configuration changes). Each request will be scoped by the vendor within 2 business days. Changes                                      |

| Support Category           | Scope Detail                                                                                                                                                                                                                                                                                                                     |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | requiring more than the remaining pool hours, or new major features, will be quoted separately.                                                                                                                                                                                                                                  |
| CMS & Technical Assistance | The vendor must provide technical assistance to DLD staff for platform operation issues that cannot be resolved independently. This includes CMS publishing issues, user access problems, and configuration queries. Up to 4 optional refresher training sessions of 1 hour each are available to DLD during the support period. |
| Infrastructure Operations  | The vendor is responsible for the ongoing operation of the hosting infrastructure, applying infrastructure-level updates, maintaining the backup and disaster recovery systems, and ensuring platform availability meets the 99.9% uptime SLA. DESC-certified SOC monitoring must remain active throughout the support period.   |

## 15.2 Support Incident SLA

| Priority      | Definition                                                                                                                                                                             | Response Time           | Resolution Time                                     | Coverage                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------|--------------------------------------------------------|
| P1 — Critical | Platform completely unavailable, or data displaying materially incorrect information on the External Portal affecting users.                                                           | Within 1 hour           | Within 4 hours                                      | 24/7/365                                               |
| P2 — High     | A core functional area is unavailable or severely degraded (AI Agent down, map not loading, exports failing, alert engine not triggering) but the rest of the platform is operational. | Within 2 hours          | Within 24 hours                                     | Business hours + on-call escalation for P1 escalations |
| P3 — Medium   | Non-critical functionality impaired, cosmetic issues, or minor data display discrepancies not affecting the accuracy of the overall published figures.                                 | Within 4 business hours | Within 5 business days                              | Business hours                                         |
| P4 — Low      | Minor usability issues, minor change requests, or operational questions that do not affect data accuracy or core functionality.                                                        | Within 1 business day   | Within 20 business days or next maintenance release | Business hours                                         |

Business hours are defined as Monday to Friday, 08:00–16:00 UAE Standard Time, excluding UAE public holidays. The vendor must provide a named support contact and an emergency out-of-hours escalation number for P1 incidents.

### 15.3 Incident Reporting

- For each P1 or P2 incident, the vendor must provide a written incident report to DLD within 3 business days of resolution, covering: incident description, root cause, resolution steps taken, and preventive measures applied.
- The vendor must maintain an incident log accessible to DLD that records all incidents raised, their priority, status, response time, resolution time, and resolution summary. DLD may request access to this log at any time.
- Any SLA breach must be proactively communicated to DLD with a root cause and corrective action before DLD raises it.

### 15.4 Warranty & Support Cost Pricing

| #                                                                    | 12-Month Warranty & Support Cost Item                                                                                                                                               | Annual Cost (AED) |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| S.1                                                                  | Dedicated support team — personnel cost and allocation. Vendor must specify the support team structure, the named primary contact, and their on-call arrangements for P1 incidents. | _____             |
| S.2                                                                  | 120 person-hours minor change pool (included in the support contract). Rate for additional hours if the pool is exhausted: AED _____ per person-hour.                               | Included          |
| S.3                                                                  | Security vulnerability monitoring and patch management activities.                                                                                                                  | _____             |
| S.4                                                                  | Infrastructure operations — hosting, backups, disaster recovery management, and uptime SLA maintenance (as applicable to the proposed hosting model).                               | _____             |
| S.5                                                                  | Help desk and incident management tooling.                                                                                                                                          | _____             |
| <b>TOTAL 12-MONTH WARRANTY &amp; SUPPORT COST — Exclusive of VAT</b> |                                                                                                                                                                                     |                   |
|                                                                      | TOTAL:                                                                                                                                                                              | _____             |

## 16. Vendor Qualifications

### 16.1 General Qualification Requirements

DLD does not impose fixed minimum years of experience, specific team sizes, or mandatory certifications beyond those related to security compliance. The following qualifications are required and will be evaluated as part of the 70% technical score. DLD encourages proposals from vendors of all sizes and backgrounds who can demonstrate genuine capability to deliver this platform.

| Qualification Area                               | Evidence Required in Proposal                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevant Experience in Digital Platform Delivery | Portfolio of completed projects demonstrating experience in at least two of the following areas: government or public sector digital platforms, real estate data platforms, PropTech applications, smart city data portals, or large-scale data visualisation systems. Project profiles must include: client name, project scope, technology stack used, and measurable outcomes. |
| Data Integration & Accuracy Track Record         | At least one case study specifically describing a data integration and accuracy validation engagement — documenting the methodology used, data quality issues discovered, and accuracy outcomes achieved. This is a mandatory case study given the critical importance of data accuracy in this project.                                                                          |
| AI/ML Delivery Capability                        | Evidence of having delivered at least one operational AI or LLM-powered application. A live demo link or recorded walkthrough (minimum 10 minutes) of a previously delivered conversational AI system is required. Arabic language capability demonstration is strongly preferred.                                                                                                |
| Security Compliance Capability                   | Active ISO 27001 certification or a DESC ISR compliance certification for the delivering organisation, or equivalent demonstrable information security management maturity. Vendor must describe their approach to DESC ISR v3 compliance and VAPT management.                                                                                                                    |
| UAE Market Presence                              | The vendor must have either a registered UAE legal entity or a clearly defined UAE-based delivery team assigned to this project. Remote-only delivery without UAE-based team members will be evaluated accordingly.                                                                                                                                                               |
| Post-Delivery Support Capability                 | The vendor must demonstrate their ability to provide the P1 support SLA (1-hour response, 24/7/365) for Critical incidents. Evidence may include: existing support agreements with similar clients, on-call team structure, and incident management tooling.                                                                                                                      |

## 16.2 Mandatory Proposal Contents Checklist

All of the following must be included in the vendor's submitted proposal. Incomplete proposals may be disqualified.

56. Company Profile: corporate overview, legal registration details in the UAE (or details of UAE-based team commitment), and a brief statement of the company's mission and relevant specialisation.
57. Portfolio of Relevant Projects: project profiles demonstrating relevant experience as described in Section 16.1. Minimum 2 project profiles; additional profiles are encouraged.
58. Data Integration & Accuracy Case Study: minimum 1 dedicated case study as specified in Section 16.1.
59. AI System Demonstration: live demo link or recorded walkthrough of an operational AI system previously delivered.
60. Proposed Technology Stack: fully detailed technology proposal for all architectural layers in Section 11.1, with justification for each choice and an explanation of how it meets the requirements of this RFP. Vendors are strongly encouraged to include an Architecture Integration Map showing how their proposed solution aligns with DLD's existing Microsoft Fabric environment (see Section 11.4). Where the vendor proposes components outside the existing ecosystem, these should be identified and justified. For the External Public Portal visualisation layer, vendors should include the comparative analysis described in Section 11.4.2.
61. Proposed Hosting Model(s): full description of the hosting option(s) proposed. If multiple options are offered, a clear comparison of their cost, performance, and risk profile must be provided.
62. Proposed Project Team: overview of the team proposed for this engagement — roles, relevant experience, and proposed allocation. Named individuals are preferred but not mandatory at proposal stage; DLD may request named resources during evaluation.
63. Phased Delivery Timeline: the vendor's proposed timeline for each of the four phases, presented as a high-level Gantt chart or equivalent. Individual task-level timelines are not required.
64. DLD Data Familiarisation Plan: description of how the vendor will conduct the mandatory data familiarisation activities in Section 12.1, including the resources assigned and the proposed approach.
65. System Architecture Diagram: full technical architecture diagram showing all proposed components, data flows, integration points, and security boundaries for the complete platform.
66. Security & Compliance Plan: detailed plan for DESC ISR v3 compliance, VAPT schedule, SOC provider selection, and ISO 27001 roadmap.
67. Proposed SLA Document: complete SLA covering both development warranty and 12-month post-go-live support — P1–P4 definitions, response and resolution times, escalation contacts, and any penalty or credit provisions for SLA breach.
68. Itemised Cost Tables: complete tables as per Sections 14.1 (Infrastructure Setup CAPEX), 14.2 (Licence & Subscription costs), 14.3 (Development & Implementation CAPEX by phase), and 14.4 (Annual Recurring OPEX). Plus the 12-month support cost table as per Section 15.4. All prices in AED, exclusive of VAT.
69. Risk Register: project risk register identifying the principal delivery, technical, data, security, and operational risks with probability and impact ratings and proposed mitigation actions.

70. Knowledge Transfer & Exit Plan: description of how DLD will receive full platform ownership — source code repositories, IaC scripts, all documentation, and infrastructure control — at project completion and upon contract expiry.

## 17. Acceptance Criteria

### 17.1 Phase-Level Acceptance

Each phase is accepted independently. Phase acceptance requires the following to be satisfied:

- All deliverables listed for that phase in Section 13 are completed and operational in the production environment.
- All functional requirements applicable to that phase pass a structured UAT session with DLD stakeholders and end-users.
- All performance SLAs in Section 10.1 are met under load.
- The data accuracy acceptance criterion is met (Section 17.2).
- No open Critical or High-severity defects remain at the time of phase sign-off.
- The DLD Project Sponsor signs a phase acceptance form confirming sign-off.

### 17.2 Data Accuracy Acceptance Criterion

For Phase 1 and for any subsequent phase that introduces new data categories, the following data accuracy criterion must be met before phase acceptance:

- A random sample of 500 transaction records displayed on the External Portal is verified by the DLD data team against the authoritative DLD source database.
- The accuracy rate (percentage of fields matching the source record) must be 99.5% or above.
- The Pre-Publication Analytics Assessment report must be signed off by the DLD Data Liaison Officer before the phase goes live with public data.

### 17.3 Universal UAT Requirements (All Phases)

| # | UAT Test Category                 | Pass Criteria                                                                                                                                                                               |
|---|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Functional Testing                | 100% of functional requirements applicable to the phase implemented and verified. Zero Critical or High-severity open defects at phase sign-off.                                            |
| 2 | Performance Testing               | All P95 response time targets met under the concurrent user loads specified in Section 10.1.                                                                                                |
| 3 | AI Agent Accuracy (Phase 1 and 2) | Agent tested against a standardised 100-question DLD data test set. Accuracy greater than or equal to 90%. Zero instances of fabricated data. Zero instances of investment advice provided. |
| 4 | Security Testing (Phase 3)        | DESC-authorised VAPT report with zero open High or Critical findings. OWASP security scan: zero critical issues.                                                                            |
| 5 | Accessibility Testing             | Automated accessibility scan: zero Critical violations. WCAG 2.1 Level AA verified across all pages.                                                                                        |



| #  | UAT Test Category           | Pass Criteria                                                                                                                                                                                                              |
|----|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | Language & RTL Testing      | English and Arabic fully correct in Phases 1–3. All 8 languages correct in Phase 4. RTL layout verified for Arabic and Urdu. No truncated or overlapping text.                                                             |
| 7  | Data Accuracy Verification  | DLD data team verifies 500 transaction records against source. Accuracy rate 99.5% or above.                                                                                                                               |
| 8  | End-User Usability Testing  | Structured usability session with a representative group of DLD staff and investor users. System Usability Scale (SUS) score greater than or equal to 75. All critical task completion rates greater than or equal to 90%. |
| 9  | CMS Non-Technical Edit Test | DLD non-technical staff members successfully publish a content change within 10 minutes without vendor assistance.                                                                                                         |
| 10 | Alert Delivery Testing      | Test alerts triggered across all applicable alert levels and types. All delivered within specified SLA times to correct recipients.                                                                                        |

## 18. Proposal Evaluation Criteria

Evaluation Weight: 70% Technical Requirements | 30% Budget & Total Cost of Ownership Proposals that do not meet the minimum technical score threshold will not proceed to financial evaluation. The highest-ranked proposal on the combined weighted score will be recommended for contract award, subject to DLD's internal governance approvals.

### 18.1 Technical Evaluation (70% Total)

| Technical Evaluation Criterion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Weight (of 70%) | Evaluation Method                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------|
| Proposed Technical Solution: quality and appropriateness of the proposed technology stack, architecture design, hosting model, AI model approach, scalability strategy, and overall alignment with the functional and non-functional requirements of this RFP. This criterion specifically includes evaluation of how well the vendor's proposed architecture aligns with DLD's existing Microsoft Fabric ecosystem (Section 11.4) — proposals demonstrating strong alignment and a clear Architecture Integration Map will be scored more favourably. For the External Public Portal, the quality of the visualisation layer comparative analysis (Section 11.4.2) will also be evaluated. | 25%             | Technical review panel evaluation + Architecture Q&A session with the vendor's technical team |
| Data Understanding & Accuracy Methodology: credibility, depth, and practicality of the proposed DLD data familiarisation plan, analytics assessment methodology, and ongoing data accuracy monitoring framework. Given the critical importance of data accuracy to this project, this criterion carries significant weight.                                                                                                                                                                                                                                                                                                                                                                 | 20%             | Data methodology review panel + evaluation of the mandatory data integration case study       |
| Security & Compliance Approach: completeness and credibility of the security plan including DESC ISR v3 compliance roadmap, VAPT scheduling, SOC provider selection, and ISO 27001 certification pathway.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10%             | Security review panel evaluation                                                              |
| Relevant Experience & Demonstrated Capability: quality of the project portfolio, relevance of prior work to this RFP's scope, and strength of the AI system demonstration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10%             | Portfolio review + reference checks (minimum 2 client references contacted by DLD)            |
| Warranty & Technical Support Plan: credibility and completeness of the 12-month warranty offer, SLA commitments, support team structure, data accuracy monitoring plan, and AI Agent knowledge update approach.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5%              | Support plan review panel                                                                     |
| <b>TOTAL TECHNICAL SCORE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                                                                                               |
| Minimum threshold to proceed to financial evaluation: 70 out of 100 on the combined technical sub-criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 70%             |                                                                                               |

## 18.2 Financial Evaluation (30% Total)

| Financial Evaluation Criterion                                                                                                                                                                                                                                                                                                                                 | Weight (of 30%) | Evaluation Method                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------|
| Total Cost of Ownership (5-Year): the combined total of infrastructure setup CAPEX (Section 14.1), licence and subscription costs (Section 14.2), implementation CAPEX across all four phases (Section 14.3), and five years of annual recurring OPEX (Section 14.4). The proposal offering the best value for money on a 5-year TCO basis will score highest. | 20%             | Financial evaluation committee using a standard 5-year TCO model applied consistently across all proposals |
| Pricing Transparency & Completeness: the degree to which all cost tables are fully and accurately completed with no missing items, bundling of line items, or ambiguous entries. Proposals with clearly itemised, justified costs will score higher than those with opaque or bundled pricing.                                                                 | 10%             | Financial review of all four cost tables in the proposal                                                   |
| <b>TOTAL FINANCIAL SCORE</b>                                                                                                                                                                                                                                                                                                                                   |                 |                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                | 30%             |                                                                                                            |

## 18.3 Evaluation Process

71. All submitted proposals are reviewed for completeness. Incomplete proposals may be disqualified at this stage.
72. The technical evaluation panel scores each proposal against the technical criteria independently.
73. Proposals achieving 70 or above out of 100 on the combined technical score proceed to financial evaluation.
74. The financial evaluation committee reviews all qualifying proposals against the financial criteria.
75. Combined scores (70% technical + 30% financial) are calculated. The highest-combined-score proposal is recommended for contract award.
76. DLD reserves the right to request clarifications, additional information, or a presentation/demonstration from shortlisted vendors prior to final recommendation.
77. DLD reserves the right to not award the contract to the lowest-cost proposal if the technical quality is assessed as insufficient to deliver the project successfully.

## 19. Data Governance, Privacy & Ownership

### 19.1 Data Ownership

- All data processed, stored, or generated within the IRETP platform is the sole and exclusive property of Dubai Land Department.
- The vendor has no rights to use any DLD data for any purpose other than delivering the contracted services.
- Upon contract expiration, termination, or handover, the vendor must deliver all data in machine-readable, open formats within 30 calendar days and provide written certification of complete data destruction from all vendor-controlled systems within 60 days.

### 19.2 Personal Data Protection Compliance

- The system must fully comply with UAE Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data (PDPL).
- No personally identifiable information (PII) — names, Emirates ID, nationality, or contact details — may be displayed in any public-facing interface.
- All transaction data displayed publicly must be either aggregated (zone/period level) or anonymised at the individual transaction level.
- A Data Processing Agreement (DPA) must be signed between DLD and the vendor before any personal data is accessed.
- User consent management for newsletters, alert emails, and cross-session AI memory must comply with PDPL.

### 19.3 Data Quality Standards

| Data Quality Dimension | Minimum Standard                                                                                                | Measurement Frequency                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Accuracy               | 99.5% or above of all data values match the DLD source system record                                            | DLD data team audit — frequency and sample size determined by DLD |
| Completeness           | 100% of mandatory fields populated for all transactions in the last 24 months                                   | DLD data team check — frequency determined by DLD                 |
| Timeliness             | Transaction data lag must not exceed 24 hours from DLD source.<br>Dashboard KPIs must not exceed 15 minutes.    | Vendor pipeline monitoring with breach alerts to DLD              |
| Consistency            | Data values consistent across all platform modules — no discrepancy between map data and transaction table data | DLD data team reconciliation — frequency determined by DLD        |

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## 20. Additional Recommended Features

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The following features are recommended for inclusion in the platform to maximise GRETI score improvement. Vendors may propose these as part of Phase 4 or as optional additions, with separate costing where applicable:

- International Market Benchmarking (Phase 4 — included): comparison of Dubai's key metrics against 5 global peer cities (London, Singapore, New York, Paris, Hong Kong), sourced from publicly available official data and updated quarterly.
- ESG / Sustainability Module (Phase 4 — included): LEED, Estidama Pearl, and BREEAM certification status for applicable buildings and projects. The Sustainability sub-index showed the strongest improvement in GRETI 2024 and is a high-impact area for UAE ranking advancement.
- Open Data API Portal (Phase 1 — included): self-service API key registration for researchers, financial institutions, and PropTech developers. OpenAPI 3.0 documentation with interactive testing console.
- Public Developer Scorecard (Phase 3 — included): simplified public-facing version of the developer rating displaying completed projects, on-time delivery percentage, RERA compliance rating, and total units delivered.
- Beneficial Ownership Transparency: display of publicly available corporate ownership structures for registered developers. Specifically cited in JLL's 2024 GRETI report as a key area for UAE improvement.
- Mortgage & Debt Market Transparency: aggregate data on mortgage financing activity — total values registered, average LTV ratios by zone, and MoM trend. GRETI 2024 identified debt market transparency as a significant gap for UAE.

## 21. Glossary of Terms

| Term / Acronym | Definition                                                                                                                                                              |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAPEX          | Capital Expenditure — one-time investment costs incurred during the project development and delivery phase.                                                             |
| DESC           | Dubai Electronic Security Center — the government authority responsible for cybersecurity regulation and certification in the Emirate of Dubai.                         |
| DESC-CSP       | DESC Cloud Service Provider Certification — mandatory certification for any cloud provider or colocation facility serving Dubai Government entities.                    |
| EWRS           | Early Warning Risk System — the internal platform module (Phase 2) that monitors risk indicators for real estate projects and developers and triggers automated alerts. |
| GRETI          | Global Real Estate Transparency Index — published biennially by JLL & LaSalle, covering 89 countries and 151 cities across 256 indicators in 6 sub-indices.             |
| GIS            | Geographic Information System — technology for capturing, storing, managing, and visualising spatially referenced data on digital maps.                                 |
| IaC            | Infrastructure as Code — the practice of managing and provisioning infrastructure through machine-readable definition files rather than manual processes.               |
| IRETP          | Integrated Real Estate Transparency Platform — the name of the system being procured through this RFP.                                                                  |
| ISR            | Information Security Regulation — the mandatory information security framework issued by DESC for all Dubai Government entities (current version: ISR v3).              |
| LLM            | Large Language Model — an AI model trained on large datasets of text capable of generating and understanding natural language.                                          |
| OPEX           | Operational Expenditure — recurring annual costs for running, maintaining, and supporting the platform after go-live.                                                   |
| PDPL           | UAE Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data — the UAE's primary data privacy law.                                                          |
| RAG            | Retrieval-Augmented Generation — an AI architecture where a language model's outputs are grounded in real-time data retrieval, reducing hallucination.                  |
| RBAC           | Role-Based Access Control — a security model where system access permissions are assigned based on defined user roles.                                                  |
| RERA           | Real Estate Regulatory Agency — the regulatory arm of Dubai Land Department responsible for licensing developers and brokers, and managing the Escrow system.           |
| RPO            | Recovery Point Objective — the maximum acceptable amount of data loss measured in time following a system failure.                                                      |
| RTO            | Recovery Time Objective — the maximum acceptable duration within which a system must be restored following a failure.                                                   |
| RTL            | Right-to-Left — the text direction used by Arabic and Urdu, requiring full layout mirroring in user interface components.                                               |
| SLA            | Service Level Agreement — a contractual commitment defining the minimum performance and availability standards the vendor must maintain.                                |

| Term / Acronym       | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TCO                  | Total Cost of Ownership — the combined cost of all CAPEX and OPEX over the evaluation period (5 years for this RFP's financial evaluation).                                                                                                                                                                                                                                                                                                                       |
| VAPT                 | Vulnerability Assessment and Penetration Testing — a security testing process that identifies and validates exploitable security vulnerabilities in a system.                                                                                                                                                                                                                                                                                                     |
| WCAG 2.1 AA          | Web Content Accessibility Guidelines 2.1, Level AA — the internationally recognised standard for digital accessibility.                                                                                                                                                                                                                                                                                                                                           |
| Microsoft Fabric     | Microsoft's unified analytics platform that integrates data engineering, data integration, data warehousing, data science, real-time analytics, and business intelligence. DLD's implementation includes OneLake, Data Factory pipelines, semantic models, and the reporting layer, all of which are mandatory components for the IRETP vendor to leverage.                                                                                                       |
| OneLake              | The single, unified data lake storage layer within Microsoft Fabric. DLD's OneLake is structured using a Lakehouse architecture with Silver (cleansed and conformed data) and Gold (curated, analytics-ready) layers. OneLake is DLD's preferred and recommended single source of truth for all IRETP data. Vendors are strongly encouraged to serve all platform visualisations and APIs from the Gold layer to maintain data consistency and avoid duplication. |
| Silver / Gold Layers | Data processing tiers within DLD's OneLake Lakehouse architecture. The Silver Layer contains cleansed, validated, and conformed data transformed from raw source system ingestion. The Gold Layer contains aggregated, business-rule-applied, analytics-ready data sets that serve semantic models and all reporting and visualisation outputs. All IRETP platform visualisations and APIs must be served from the Gold Layer.                                    |

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### END OF DOCUMENT

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